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Fate of sterols, polycyclic aromatic hydrocarbons, pharmaceuticals, ammonia and solids in single-stage anaerobic and sequential anaerobic/aerobic/anoxic sludge digestion

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ABSTRACT

Emerging contaminants (ECs), such as pharmaceuticals, sterols and polycyclic aromatic hydrocarbons (PAHs) are frequently detected in the environment. ECs are refractory, toxic, tend to bioaccumulate and have a potential to disrupt the endocrine system of living organisms. These compounds are only partially eliminated in wastewater treatment plants (WWTPs). Due to their hydrophobic nature, they tend to accumulate in sludge. However, the fate of the majority of ECs in sludge treatment processes is not fully understood. In this study, the effect of a sequential anaerobic/aerobic/anoxic (AN/AERO/ANOX) digestion and a conventional single-stage AN digestion (as control) was investigated on mixed primary and secondary sludge. Digesters were operated at an overall solid retention time (SRT) of 18 days. The steady-state results have shown that sequential AN/AERO/ANOX digestion configurations improved the removal of three classes of ECs (e.g., sterols, PAHs and pharmaceuticals) by either reducing their accumulation or enhancing their removal. Moreover, sequential AN/AERO/ANOX digestion also achieved 45% less ammonia generation, 20% faster digestate dewaterability and 4% enhanced volatile solids removal compared to single-stage AN digestion.

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1. Introduction

Over the last few decades, the occurrence of emerging contaminants (ECs), such as pharmaceuticals, sterols and polycyclic aromatic hydrocarbons (PAHs), have gained considerable attention due to their regular occurrence in the environment and their potential to cause harmful effects in the environment and living organisms. ECs are naturally occurring or synthetic chemicals that are not usually monitored in the environment. They have a potential to disrupt the endocrine system, can cause infertility as well as expedite the maturation in females and bioaccumulation in aquatic organisms (mainly fish) (Vanderford et al., 2003; Jakimska et al., 2014). Moreover, these compounds are part of many products that are consumed daily (medicines, over the counter therapeutic drugs, painkillers and industrial agents) at home or used in industrial processes. ECs are refractory compounds. Due to an increase in the consumption of ECs, their rate of introduction into the environment surpasses their degradation rate and hence, they are detected worldwide in the environment.

As one of the major groups in ECs, pharmaceuticals have a crucial role in disease treatment and prevention in both humans and animals. They encompass a wide and diverse group of chemicals designed either to be very toxic to infectious organisms or highly active to interact with receptors in humans and animals. The usage and consumption of pharmaceuticals have increased significantly because of population increase and aging, the discoveries of new drugs and the availability of less expensive generics. After ingestion, significant fractions of the parent compounds either as active/inactive metabolites or in unmetabolized form are excreted from the body and are then found in raw wastewater (Ternes et al., 2001; Celiz et al., 2009). Moreover, pharmaceuticals along with their metabolites are constantly entering wastewater via disposal of unused or expired drugs or directly from pharmaceutical discharges (Gros et al., 2006).

Another set of compounds, known as sterols, are also recurrently detected in wastewater treatment plants (WWTPs). A major fraction of municipal wastewater consists of organic matter, 20–25% of which is composed of lipids (mainly cholesterol) (Mahmood-Khan, 2005). According to raw wastewater analysis, 64% of the lipids in municipal wastewater are from fecal sources (Mahmood-Khan, 2005). Coprostanol accounts for 40–60% of the total fecal sterols excreted in human wastes and has been

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extensively used as an indicator of fecal contamination. Coprostanol is formed in the human digestive track by hydrogenation of cholesterol (Bujagić et al., 2016). However, degradation of cholesterol in the environment leads to the formation of cholestanol (Bujagić et al., 2016). Both cholestanol and coprostanol are isomers of each other.

PAHs are another class of ECs found in the environment. They are organic compounds that contain two or more benzene rings. PAHs containing up to six aromatic rings are known as small PAHs while others are called large PAHs (Abdel-Shafy and Mansour, 2016). Most of the research has been conducted on small PAHs due to their availability (Abdel-Shafy and Mansour, 2016). They are colorless, white or pale-yellow solids (Abdel-Shafy and Mansour, 2016). PAHs can be formed because of incomplete combustion from either natural combustion sources (forest fires) or anthropogenic sources such as automobile emission and fuel reprocessing industries (pyrolysis). Due to their hydrophobic nature, PAHs are removed during biological wastewater treatment from aqueous phase via adsorption to solids and accumulate in sludge (Poluszyńska et al., 2017). They also tend to bioaccumulate in living organisms (Liu et al., 2017), and as potent immune suppressants they can cause cancer as well as mutations (Abdel-Shafy and Mansour, 2016).

Depending on the treatment processes used at the industrial or municipal WWTPs, the removal efficiency of ECs changes. Furtula et al. (2011) studied removal of sterols from six WWTPs treating mostly domestic wastewater. Out of six WWTPs studies, three of them had secondary treatment while other three have tertiary treatment. He found that in general, all six WWTPs were effective in removing sterols (Furtula et al., 2011). However, tertiary WWTPs showed slightly better sterols removal efficiencies compared to secondary WWTPs (Furtula et al., 2011). Similarly, Jelic et al. (2012), studied removal of pharmaceuticals in WWTPs. Jelic concluded that some pharmaceutical such as ibuprofen generally showed high removal (>75%) while others such as diclofenac demonstrated variable removal efficiency (0–90%) (Jelic et al., 2012). On the other hand, PAHs with up to three aromatic rings (also known as low molecular weight PAHs) showed high removal (33–100%) during biological treatment process compared to PAH with four or more aromatic rings (also known as high molecular weight PAHs) (Qiao et al., 2014).

Fate and removal of ECs in WWTPs are complex and difficult to assess due to tedious and cost-intensive analyses. In addition, there are a very few studies reporting the fate of these compounds in wastewater sludge treatment. To the best of the authors' knowledge, no study investigated the impact of sequential anaerobic/aerobic/anoxic (AN/AERO/ANOX) sludge digestion on the fate of the aforementioned classes of ECs.

In this research, the fate of four pharmaceuticals, two PAHs and three sterols in single-stage AN and sequential AN/AERO/ANOX digestion was compared. A secondary wastewater treatment facility was chosen to serve as the test model; however, the proposed approach could also be applicable to other treatment facilities. The selected wastewater treatment facility employs primary and secondary wastewater treatment. Primary sludge is produced through primary sedimentation, which occurs after screening and pre-aeration grit removal. The secondary treatment process involves soluble biochemical oxygen demand removal in trickling filters, floc formation in solids contact aeration and solids/liquid separation in secondary clarification. The effluent is disinfected by chlorine before discharge. At the treatment plant, primary sludge is gravity thickened to a target of 4.5% solids, while secondary sludge goes through a dissolved air flotation process with a target of about 5% solids. The waste sludges are mixed together (comprised of about 55% primary, 45% secondary by volume) before being fed into four parallel thermophilic digesters operated at 55°C to generate Class A biosolids and the dewatering centrate is returned to the headworks. Due to the high cost associated with both the construction of additional digestion capacity and the ultimate disposal of biosolids, the wastewater treatment facility was interested in advanced solids digestion technologies that can minimize solids, extend the useful life of existing digesters and offer additional benefits. The reason for selecting AERO/ANOX process to polish AN digester effluent was due to the benefits (e.g. lower ammonia generation, reduced odor for dewatered biosolids and improved dewaterability/pathogen destruction) provided by AERO process while keeping the aeration cost low with the help of ANOX process. Also, the use AERO/ANOX process was selected to see the impact of diverse microbial community on the removal of emerging contaminants.

Physico-chemical properties are important in the prediction of the fate of ECs in the environment. The physico-chemical properties of the ECs studied are summarized in Table 1. Among the

Table 1
Physico-chemical properties of selected emerging contaminants.

Class of compounds	Chemical Name	Molecular Formula	Molecular Weight (g/mole)	log K_{ow}[a]	pK_a[b]	Henry's law constant (atm-m^3/mol)	Solubility in water[c] (mg/L)
Pharmaceuticals	Diclofenac	$C_{14}H_{10}Cl_2NO_2$	296.16	4.5[d]	4.1[d]	$5.3 \times E^{-9}$[e]	2.4[e]
	Ibuprofen	$C_{13}H_{18}O_2$	206.23	4.0[d]	4.9[d]	$1.2 \times E^{-6}$[e]	21.0[g]
	Fenofibric Acid	$C_{17}H_{15}ClO_4$	318.75	4.0[e]	2.9[h]	$7.9 \times E^{-10}$[e]	–
	Trazodone	$C_{19}H_{22}ClN_5O$	371.87	3.2[f]	6.1[f]	–	27.6[f]
PAHs*	Biphenyl	$C_{12}H_{10}$	154.21	4.0[f]	–	–	–
	Phenanthrene	$C_{14}H_{10}$	178.23	4.5[f]	–	$4.2 \times E^{-5}$[f]	1.1[f]
Sterols	Cholesterol	$C_{27}H_{46}O$	386.66	8.7[f]	15.0[i]	$1.7 \times E^{-4}$[f]	–
	Cholestanol	$C_{27}H_{48}O$	388.68	8.8[k]	6.5[k]	$1.6 \times E^{-4}$[k]	–
	Coprostanol	$C_{27}H_{48}O$	388.68	8.8[j]	6.5[j]	–	–

* PAH: Polycyclic aromatic hydrocarbons.
[a] Octanol-water partition coefficient (log K_{ow}).
[b] Ionization constant (pK_a).
[c] Solubility in water in mg/L at 25°C.
[d] Gros et al. (2006).
[e] Muñoz et al. (2008)
[f] pubchem.ncbi.nlm.nih.gov.
[g] Suárez et al. (2008).
[h] Rosal et al. (2010).
[i] Hamilton Giles et al. (2012).
[j] Vigneswaran (2009).
[k] www.chemspider.com.

pharmaceuticals studied in this work, diclofenac and ibuprofen are endocrine disrupting compounds (David and Pancharatna, 2009; van den Brandhof and Montforts, 2010). Most of the ECs have high octanol-water partition coefficient values ($\log K_{ow} > 3$) and they tend to partition into the sludge during wastewater treatment (Gros et al., 2006; Poluszyńska et al., 2017). Another commonly shared feature among ECs listed in Table 1 is a relatively low Henry's law constant ($<10^{-4}$) and therefore preferential partitioning into the water instead of air (Rogers, 1996).

2. Digester set-up

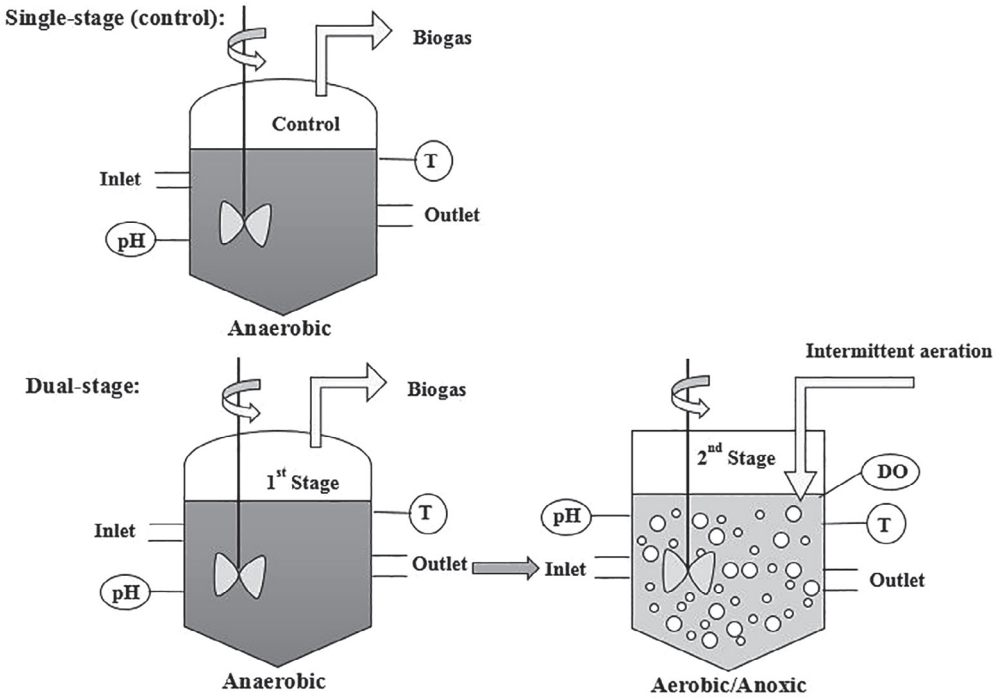
Fig. 1 illustrates the digester set-up. One AN digester served as a single-stage control digester while the other was used as a 1st stage in the sequential AN/AERO/ANOX system. Both AN digesters had a working volume of 5.5 L.

Mixed sludge containing 55% primary and 45% secondary sludge from the secondary treatment stage of a municipal WWTP (BC, Canada) was used to feed the AN digesters. The characteristics of mixed sludge used are given in supplementary material (Table S1).

The 2nd stage digester in the sequential AN/AERO/ANOX system was operated with intermittent aeration to establish AERO/ANOX conditions. A thermal mass flow controller regulated the airflow to the 2nd stage AERO/ANOX digester. AERO conditions were achieved by keeping the airflow on and ANOX conditions were established by turning off the airflow into the 2nd stage digester. All digesters were operated at thermophilic conditions (55°C). The working volume of the 2nd stage AERO/ANOX digester was 0.95 L. Thermophilic anaerobic inoculum from the same WWTP as the mixed sludge was used to start up the two AN digesters. The characteristics of the anaerobic inoculum are provided in supplementary material (Table S2).

The AERO/ANOX vessel was initially inoculated with activated sludge from the same facility. The overall sludge retention time (SRT) was kept constant at 18 d for both single-stage and sequential digestion process to compare performance results and correspond to the sludge age of the full-scale thermophilic single-stage digesters at the municipal WWTP. The digesters were operated under two different scenarios. In Scenario 1, for sequential AN/AERO/ANOX digesters, the 1st stage AN and the 2nd stage AERO/ANOX digesters were operated at an SRT of 15 and 3 d respectively, hence making the overall SRT of the system equal to 18 d. In Scenario 2, the SRT of the 1st stage AN digester marginally increased to 16.5 d while the SRT of the 2nd stage AERO/ANOX digester was decreased to 1.5 d, keeping the overall SRT of the system at 18 d. The short SRT for AERO/ANOX digesters and high digester temperature/pH (55°C/8.0) allowed to minimize energy required for aeration and stripping high ammonia concentration coming from 1st stage AN digester, respectively. Average dissolved oxygen (DO) was maintained around 2 mg/L and below 0.2 mg/L for AERO and ANOX phases, respectively. The summary of digestion scenarios for automated digesters is given in Table 2.

A New Brunswick BioFlo® 115 control station and fermenter glass vessels were used for digester setups. One main control center, capable of operating two additional utility control stations, was used to operate three automated digesters as given in Fig. 1. The main control center was equipped with Biocommand software to control the operation of all three digesters. Each glass vessel was equipped with a heat blanket or a water jacket to maintain the setpoint temperature. Moreover, each fermenter vessel had DO, pH and temperature probes to monitor the digester operation. Peristaltic pumps attached to each control station were used to provide continuous automated feeding and wasting. Pitch blade mixers provided continuous mixing in automated digesters. Automated digesters were operated for a total of 212 days for Scenario 1 and 145 days for Scenario 2. Each AN digester was at steady-state



Solid retention time (SRT) for single-stage digester = 18 d
Scenario 1- SRTs: 1st stage digester = 15 d, 2nd stage digester = 3 d, Overall SRT = 18 d
Scenario 1- SRTs: 1st stage digester = 16.5 d, 2nd stage digester = 1.5 d, Overall SRT = 18 d

Fig. 1. Digester configurations used in experimental set up.

Table 2
Summary of the digestion scenarios used for automated digesters.

SCN[a] No.	Single/Dual Stage	Anaerobic	Aerobic/Anoxic		Total SRT (days)
		SRT[*] (days)	SRT[*] (days)	Aeration cycle (hours on:off)	
1	Single-stage[†]	18	–		18
	Dual-stage	15	3	1:1	18
2	Single-stage[†]	18	–		18
	Dual-stage	16.5	1.5	1:1	18

[a] SCN: scenario.
[†] Single-stage digesters were operated as "control" and they simulated existing full-scale digesters.
[*] SRT: solids retention time.

for 190 days in Scenario 1 and 120 days in Scenario 2, respectively. Steady-state corresponded to a period during which variation in daily biogas production was less than 10%.

3. Analytical methods

3.1. Conventional pollutants

Several conventional parameters were analyzed to assess digester performance. These parameters included biogas quantity and composition, headspace volatile sulfur compounds (VSCs), solids removal, chemical oxygen demand (COD), alkalinity, pH, fecal coliforms, total volatile fatty acids (TVFAs), ammonia, nitrate, nitrite, ortho-phosphate and capillary suction time (CST).

Biogas samples from the headspace of AN digesters were analyzed for biogas composition by using a method proposed by van Huyssteen (1967). An Agilent 7820A gas chromatograph (GC) equipped with a thermal conductivity detector and packed column (Agilent G3591-8003/80002) was used to determine biogas percentage composition in terms of CH_4, CO_2, O_2 and N_2. Daily volume of biogas produced was measured by using two Ritter milligas counters at room temperature and then converted into standard temperature and pressure (STP) of 0 °C and 1 atm.

Organics analysis were carried out by analyzing samples for total solids (TS) and volatile solids (VS) using Standard Methods procedures 2540 B and 2540 E, respectively (APHA, 2005). Moreover, chemical oxygen demand (COD) was determined by using the closed reflux calorimetric method (Standard Methods 5220 D, APHA, 2005) with Thermo Scientific™ GENESYS™ 10S UV–Vis spectrophotometer (Thermo Fisher Scientific, Inc., Ottawa). Alkalinity analysis was carried out by using the titration method described in Standard Methods 2320 B (APHA, 2005). Nitrogen species such as ammonia, nitrite and nitrate were measured by using 4500-NH_3 D, 4500-NO_2-calorimetric method and 4500-NO_3-nitrate electrode method, respectively (APHA, 2005). The rate of dewaterability of digester effluent was determined using Standard Methods 2710 G. TVFAs, which constitute the summation of acetic, propionic and butyric acid, were analyzed by using GC equipped with flame ionization detector. TVSCs, comprised of hydrogen sulfide, methyl mercaptan, ethyl mercaptan, dimethyl sulfide, carbon disulfide, ethyl methyl sulfide, 1- propanethiol and dimethyl disulfide, were analyzed using GC equipped with a sulfur chemiluminescence detector. The ortho-phosphate (ortho-P) analysis was carried out by using the Ascorbic Acid Method 4500-P E (APHA, 2005). This method uses ammonium molybdate to react with antimony potassium tartrate in an acid medium with ortho-P to produce a heteropoly acid-phosphomolybdic acid that is reduced to deep blue colored molybdenum blue by the addition of ascorbic acid (APHA, 2005).

3.2. Analytical methods for emerging contaminants

Although there are numerous ECs present in sludge streams, 211 ECs belonging to the group of alkylphenols, hormones, sterols, PAHs, pesticides and pharmaceuticals were analyzed for three independent batches of samples. Each batch consisted of digester feed (mixed sludge) and effluents from the single-stage (control) and the two dual-stage digesters. Among the ECs analyzed, only nine were quantified successfully in all four samples (no spiking) and were comprised of three sterols, two PAHs and four pharmaceuticals.

3.2.1. Pharmaceuticals

There were four pharmaceuticals analyzed in this research. Ibuprofen was analyzed by using AXYs method MLA-075, whereas the remaining three pharmaceuticals, diclofenac, trazodone and fenofibric acid, were analyzed by AXYS method MLA-104 (AXYS, 2016b, 2016c). MLA-075 is a revised version of US EPA 1694 method, which was also developed by AXYS Analytical Services (BC, Canada) (USEPA, 2007b). Methods MLA-075 and MLA-104 are suitable for the analysis of various hormones, pharmaceuticals and personal care products in solid, aqueous and polar organic compound integrative samples. The process of sample extraction, cleaning and analysis procedures using liquid chromatography-tandem mass spectrometry (LC-MS/MS) are identical in both MLA-075 and MLA-104, however, each method targets a different suite of pharmaceuticals. A summary of the methodology used in MLA-075 and MLA104 is given in the section below.

According to MLA-075 and MLA104, the analysis requires an extraction of 0.25 g of sludge samples at two different pH conditions depending on the acidic or basic nature of compounds. All four pharmaceuticals studied are weak acids so their analyses were carried out at acidic pH (pH = 2). Each sample was buffered to pH 2.0 using phosphate buffer and spiked with a suite of isotopically labeled internal (surrogate) standards, which included $^{13}C_6$-ibuprofen, $^{13}C_6$-diclofenac, trazodone-d_6 and fenofibric acid-d_6. Samples were extracted using sonication with aqueous buffer/acetonitrile (two extractions) and finally with acetonitrile. The combined extracts were concentrated to evaporate the acetonitrile, diluted to 200 mL with ultra-pure water and cleaned up by solid phase extraction using Waters' Oasis hydrophilic-lipophilic balance (HLB) cartridges. Cleaned samples were then analyzed by LC-MS/MS. A recovery standard was used to monitor the recovery of the isotopically labeled standards before extraction. The LC-MS/MS was ran in the multiple reaction monitoring mode (MRM) using electrospray ionization (ESI) either in the positive mode (for diclofenac and trazodone) or in the negative mode (for ibuprofen and fenofibric acid). Separate analysis runs and conditions were used for each of the four compounds. At least one parent to daughter transition was monitored for each analyte and for each isotopically labeled standard. All analytes were quantified by isotope dilution internal standard quantification which compares the response of each target analyte to that of its isotopically labeled analog. This procedure produces accurate results that are recovery corrected for losses during the analysis process and compensated for LC-MS/MS suppression or enhancement due to sample matrix.

3.2.2. Sterols

US EPA 1698 is a method developed for the analysis of sterols and hormones in water, soil, sediments and biosolids by high-resolution gas chromatography (HRGC) and high-resolution mass spectrometry (HRMS) (US EPA, 2007a). The method consists of solvent extraction of the sample, followed by clean-up with a layered alumina/Florisil column and an option to remove sulfur using copper. After the clean-up, the target analytes are derivatized to make them volatile and thus compatible for the analysis by HRGC/HRMS (US EPA, 2007a). The techniques of isotope dilution and the use of internal standards are applied for the quantitation.

According to US EPA 1698 method, for sludge analysis, 0.25 g of sample was spiked with isotopically labeled compounds before extraction. Each sample was extracted by sonication with aqueous buffered (pH 2.0–3.5) methanol. The extracts were diluted to 200 mL with ultra-pure water and cleaned by solid phase extraction (SPE) on Waters' Oasis HLB (Hydrophilic/Lipophilic Balanced) cartridges. The extracts were then dried and derivatized with BSTFA (N,O-Bistrifluoroacetamide), and spiked with recovery standard (pyridine-d_{10}) prior to analysis by capillary gas chromatography and high resolution mass spectrometry (GC/HRMS), which was operated in the multiple ion detection (MID) mode and acquired at least two characteristic ions for each analyte and internal standard. For degradation products of cholestereol i.e. coprostanol and cholestanol which are isomers and have same molecular weight, their quantification ions have different *m*/*z* ratios. For instance, coprostanol have m1 = 370.36 and m2 = 371.37. However, cholestanol has m1 = 445.39 and m2 = 446.39. Each individual sterol was identified by comparing its GC retention time and the ion abundance ratios of the two *m*/*z* ions to those of the analyte in the calibration standard. A compound was identified when the retention times and ratios agree. The quantitative analysis of the sterols was performed by internal standard quantification using the cholesterol-d_7 as the quantification reference.

3.2.3. Polycyclic hydrocarbons

Biphenyl and C2-Phenanthrenes/Anthracenes were the two PAH compounds studied in this work. These compounds are analysed by AXYs method MLA-021 (AXYS, 2016a). In this method, sludge samples (2 g dry weight equivalent) were spiked with deuterated PAH internal standards (biphenyl-d_{10} and phenanthrene-d_{10}) dried by mixing with anhydrous sodium sulfate and soxhlet extracted with toluene. The extract was concentrated to 1 mL and cleaned by adsorption column chromatography on alumina. An aliquot of deuterated recovery standard was added and the extract was analyzed by gas chromatography with mass spectrometric detection (GC/MS). The GC/MS was operated in the selected ion mode and two mass fragments were monitored for positive identification of each PAH and labeled standard. Isotope dilution internal standard quantification was used.

Calibration of ECs was performed using calibration standards (5 to 8 points depending on the analyte), each containing the analyte of interest at varying concentrations and its isotopically labeled analog at a fixed concentration. The entire set of calibration standards was ran at the beginning of each analytical series and a mid-level calibration standard was analyzed time to time to verify calibration.

4. Statistical analysis

Statistical analysis comprised of mean value and standard deviation was calculated for all the data. The experimental data was also analyzed using general linear model of Minitab software, to determine whether the means of two or more groups differ. General linear model is an analysis of variance (ANOVA) procedure. It uses least square regression approach in calculations to describe the statistical relationship between one or more predictors/factors and a continuous response variable. After using a best fit general linear model, multiple comparisons between different pairs of digester were carried out for significant difference analysis. Multiple comparisons were executed using Fisher's Least Significant Difference test to assess the statistical significance of differences between means using a 95% confidence interval.

5. Results and discussion

5.1. Conventional parameters

Daily biogas production from AN digesters normalized by VS added at standard temperature and pressure (STP) of 0 °C and 1 atm is presented in Fig. 2.

The average of daily specific biogas volumes ranged from 532 to 585 mL/g VS fed during the AN digester operation (Fig. 2). The amount of specific biogas produced per g of VS fed increased with the increase in SRT of the AN digester due to the low organic loading rate (OLR) associated with higher SRTs (e.g., OLR of 3.83, 3.48 and 3.19 kg COD/m^3/d at SRTs of 15, 16.5, 18 days, respectively). The control digester produced the highest amount of specific biogas (585 mL/g VS fed). The relationship between the increase in specific biogas amount per gram VS fed (Fig. 2) and SRT was found to be statistically significant ($P = 0.02 < 0.05$).

All AN digesters indicated similar headspace biogas composition with methane, carbon dioxide, nitrogen and oxygen concentrations ranging from 64 to 68%, 29 to 35%, 0.1 to 3.4% and 0.04 to 1.2%, respectively.

The average VS removal efficiencies across individual single-stage and dual-stage digesters are shown in Fig. 3. The figure shows the averages and error bars display the standard deviation for 22–34 replicates from steady-state. Overall, the sequential AN/AERO/ANOX digestion system resulted in higher VS removal than the control digester (i.e. 57.8% versus 60.3% and 58.2%) for Scenarios 1 and 2, respectively. The sequential AN/AERO/ANOX system with an aeration cycle ratio of 1-hr:1-hr in Scenarios 1 and 2 achieved 4.2 and 0.6% higher VS removal than the control digester, respectively (Fig. 3). Hence, it can be concluded that the overall VS removal of sequential AN/AERO/ANOX process increased with an increase in the SRT of the AERO/ANOX digesters, which agrees with the literature (Ahmad et al., 2016; Kim and Novak, 2011).

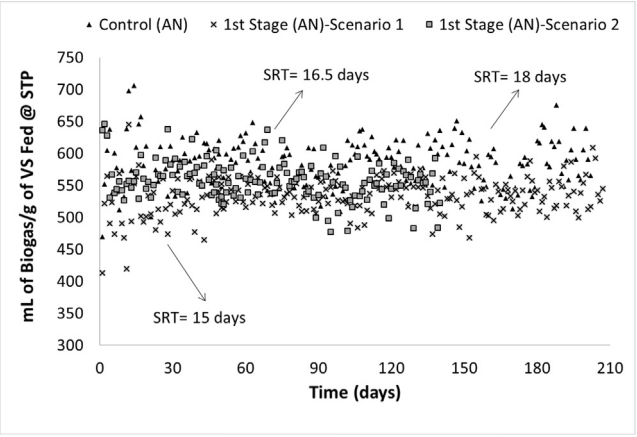


Fig. 2. Daily specific biogas production per gram of volatile solids (VS) fed with digestion time.

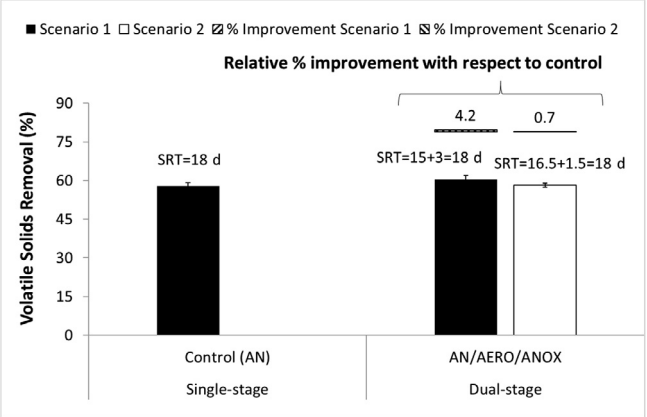


Fig. 3. Average volatile solids removal of single-stage and dual-stage digesters.

The improvement (4.2%) of VS removal offered by the sequential AERO/ANOX system may seem very low but depending on the scale of WWTP, this may translate to a significant saving in sludge disposal. For example, for a WWTP that serves a population of 1 million, this translates to $250,000 cost saving per year. Furthermore, the main advantage of the sequential sludge treatment is not VS reduction, but, additional operational benefits (discussed in the following sections), such as ammonia removal, enhancement in rate of dewaterability and pathogen destruction along with the reduction of odours.

In AN digesters, ammonia is generated from the degradation of organic nitrogen which under AERO conditions gets converted into nitrite and nitrate by oxidation with the help of *Nitrosomonas* and *Nitrobacter* bacteria, respectively (Tchobanoglous et al., 2003). Average ammonia concentrations for the mixed sludge and the effluent streams of single-stage and dual-stage digesters are given in Fig. 4. Error bars in Fig. 4 represent standard deviation of 16 to 25 replicates from steady-state. In Scenarios 1 and 2, all AN digesters on average doubled in ammonia concentration (i.e. 1327–1399 mg/L) compared to incoming mixed sludge (i.e. 662–710 mg/L). SRT had a significant effect on the concentration of ammonia generated by the AN digesters ($P = 0.00 < 0.05$). Moreover, the 2nd stage AERO/ANOX digester at an aeration cycle ratio of 1-hr:1-hr, lowered the ammonia concentration of the 1st stage AN digester back to 763 and 1003 mg/L in Scenarios 1 and 2, respectively. The decrease in ammonia concentration was also found to be significant ($P = 0.00 < 0.05$) with the increase in SRT of the 2nd stage AERO/ANOX digester. The decrease led to 44.5 and 28.3% lower ammonia generation in Scenarios 1 and 2, respectively, compared to the control digester. Nonetheless, the benefit of using AERO/ANOX digester to reduce ammonia produced by AN digester was clearly shown.

Ammonia gas reacts with water to form ammonium hydroxide ions and both exist in equilibrium depending on the pH and temperature of the solution. The higher the temperature and pH, the higher the unionized free ammonia in the solution (Metcalf & Eddy et al., 2014). Therefore, one way to remove ammonia is to have high pH and temperature condition. Due to thermophilic conditions and pH of 7.65, the high levels of ammonia produced by the 1st stage (AN) digester of the sequential AN/AERO/ANOX digestion were reduced by stripping because biological nitrification and denitrification are both inhibited at temperatures great than 35°C (Layden et al., 2007). Moreover, biological nitrification is fully halted above 40–45°C (Yuan et al., 2014). In terms of biological removal, the average nitrate concentration in the 2nd stage AERO/ANOX digester ranged from 16 mg/L to 4.2 mg/L in Scenarios 1 and 2, respectively (Table 3). In general, the concentration of nitrate increased as the SRT of AERO/ANOX digester increased. On the contrary, the concentration of nitrite in AERO/ANOX digester increased with decrease in SRT from 3 to 1.5 d for Scenarios 1 and 2, respectively. The average nitrite formed in AERO/ANOX digester varied from 0.09 mg/L to 7.13 mg/L across both scenarios (Table 3). The low average concentration of nitrite indicated that almost complete oxidation of nitrogenous species (ammonia and nitrite to nitrate) was achieved in Scenario 1. There was no recycling in the digesters. Hence, the SRT of all the digesters was equal to the hydraulic retention time (HRT) during which ammonia stripping occurred.

Dewaterability is a crucial parameter in sludge management and is directly related to the amount of conditioning agent required and affects the cost associated with biosolids disposal or reuse. Sludge dewaterability can be negatively affected by the high amount of free or bound water held by the sludge floc structure

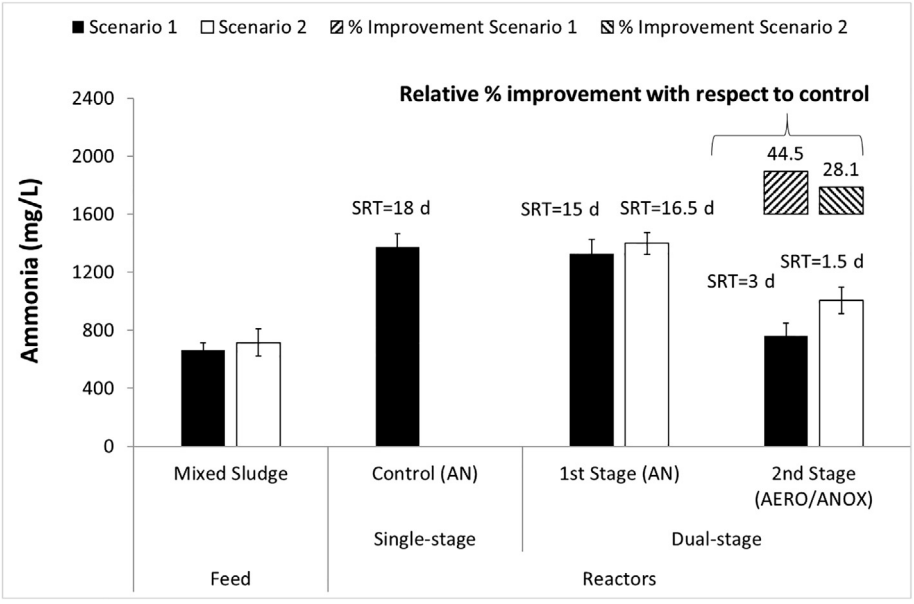


Fig. 4. Average ammonia concentration in mixed sludge feed and effluent sludge digesters.

Table 3
Steady-state results of single-stage and sequential digestion.

	Single-stage Anaerobic (AN)	Sequential Anaerobic/Aerobic/Anoxic (AN/AERO/ANOX)	
	Scenario 1		Scenario 2
Digesters	Control (AN)	AERO/ANOX (1 h:1 h)	AERO/ANOX (1 h:1 h)
COD removal (%)	58.12 (1.5; 22)[†]	61.10 (1.5; 13)	58.37 (0.7; 14)
Nitrite (mg/L)	–	0.09 (0.03; 14)	7.13 (0.9; 15)
Nitrate (mg/L)	–	16.02 (3.6; 14)	4.24 (1.1; 15)
Ortho-phosphate (mg/L)	146 (9; 24)	128 (15; 14)	133 (9; 15)
Alkalinity (mg/L as $CaCO_3$)	5228 (96; 24)	1998 (85; 14)	2740 (110; 15)
Capillary suction time (sec/%TS by wt.)	956 (49; 3)	764 (76; 3)	800 (70; 3)
Headspace volatile sulfur compounds (mg/L)[*]	105 (22; 7)	98[a] (27; 4)	95[a] (31; 3)

[†] Data represent the arithmetic mean of measurements (standard deviation; number of data points).
[*] Volatile sulfur compounds include summation of hydrogen sulfide, methyl mercaptan, ethyl mercaptan, dimethyl sulfide, carbon disulfide, ethyl methyl sulfide, 1-propanethiol and dimethyl disulfide.
[a] Data belongs to 1st stage anaerobic digester of sequential digestion system.

and extra-cellular polymeric substances (EPS) via electrostatic interactions and hydrogen bonds (Zhou et al., 2002). CST test shows the rate at which a sample releases its water content. Thus, a decline of CST value suggests the improvement of the dewaterability rate (Yu et al., 2008). The CST results normalized by dry solids content (TS% by weight) of the digester effluents are given in supplementary material as Fig. S1, where data represent averages and error bars represent standard deviations of 3 samples analyzed in triplicates from steady-state.

Novak et al. (2003) found that thermophilic temperature results in poor dewaterability compared to mesophilic whereas AN conditions generate sludge with slower dewaterability in comparison to AERO conditions. The CST of the control digestate was measured to be 956 s/TS% by weight. On the other hand, the sequential AN/AERO/ANOX digestate showed relatively enhanced dewaterability with a normalized CST value of 764 s/TS% by weight which corresponded to a 20% improvement in dewaterability compared to the control digester (Fig. S1). Due to the decrease in SRT of the 2nd stage AERO/ANOX digester, the dewaterability rate was inferior in Scenario 2, with 16% improvement compared to the single-stage digester. The rate of dewaterability was found to be significantly affected by the SRT of the AERO/ANOX digester ($P < 0.05$).

Phosphorus is a vital element for all organisms and it is a key element in fertilizers used for food production. Waste sludge resulting from municipal wastewater treatment contains significant quantities of phosphorus. This is due to the presence of phosphate accumulating organisms (PAOs), particularly in WWTP with biological nutrient removal processes. These organisms can store phosphorus as polyphosphate (Poly-P) under specific conditions (Ge et al., 2013). Under AN conditions, PAOs take up VFAs and store them as polyhydroxyalkanoates (PHAs). However, when PAOs are exposed to AERO conditions, they use their stored PHA as an energy and carbon source for their growth and take up an excessive amount of ortho-P to produce intracellular poly-P. Net P is removed from the system via wasting of excessive sludge (Ge et al., 2013). In this study, the average ortho-P concentration found in the single-stage AN digester was 146 mg/L. In sequential AN/AERO/ANOX system, incoming concentration of ortho-P (137 mg/L) from 1st stage AN digester was decreased to 128 mg/L by 2nd stage AERO/ANOX digester in Scenario 1. Similarly, in Scenario 2, 2nd stage AERO/ANOX digester decreased the incoming 140 mg/L ortho-P from 1st stage AN digester to 133 mg/L (Table 3). The decrease in ortho-P concentration due to 2nd stage AERO/ANOX digester is in accordance with the literature (Ge et al., 2013). The ANOVA analysis showed that the decrease in SRT of the AERO/ANOX digester was statistically significant with the decrease in the ortho-P removal.

Digester headspace VSCs are generated by the degradation of organic and inorganic sulfur compounds during the AN digestion of sludge. The average concentration of VSCs in the single-stage control digester was 105 mg/L as given in Table 3. However, VSCs were not detected in the headspace of the 2nd stage AERO/ANOX digester. The alkalinity produced by AN digesters ranged from 5163 to 5228 mg/L $CaCO_3$. In the 2nd stage AERO/ANOX digester, alkalinity was reduced to an average value of 1998 and 2223 mg/L for dual-stage digestion system in Scenarios 1 and 2, respectively. Furthermore, the concentration of TVFAs in AN digesters was around 70 mg/L and within the safe range of <250 mg/L (Tchobanoglous et al., 2003), which indicated that the majority of TVFAs were converted to methane within the SRT and OLR ranges used.

5.2. Emerging contaminants

In this section, details of the results on the fate of selected ECs in single-stage AN and dual-stage sequential AN/AERO/ANOX sludge digestion processes are discussed. When presenting results, positive and negative % changes in the sludge digester loading rate of ECs compounds are represented as an accumulation and a reduction in digesters, respectively, relative to feed (mixed sludge). Also, only Scenario 1 was considered for the study on the fate of ECs in sequential digestion pertaining to the higher SRT (3 d) of 2nd stage digestion (yielding higher removal in conventional parameters) compared to 1.5 d in Scenario 2. In figures reporting on EC removals, data and error bars represent the average and standard deviation respectively of three independent sampling during steady-state.

5.2.1. Pharmaceuticals

In this research, two nonsteroidal anti-inflammatory drugs (NSAIDs); ibuprofen and diclofenac were studied. As displayed in Fig. 5a, ibuprofen showed 62% accumulation in the single-stage digester compared to 12% accumulation in the dual-stage system because it is degradable under AERO conditions (Huang et al., 2011). Ibuprofen in general displays a high removal rate in WWTPs (more than 80%) which can be linked mainly to biodegradation in sludge (Jelic et al., 2011; Langenhoff et al., 2013). However, previous studies have also reported persistence of ibuprofen in AN conditions (Alvarino et al., 2014; Wijekoon et al., 2015; Yang et al., 2017). According to Alvarino et al. (2014), the branched substitutions on para-position of the aromatic ring of ibuprofen creates resistant to degradation under AN conditions. On the other hand, ibuprofen is highly biodegradable in aerobic conditions. Thus, the accumulation of ibuprofen in sequential AN/AERO/ANOX system is lower (12% vs. 62%) than single-stage AN digestion. Although lit-

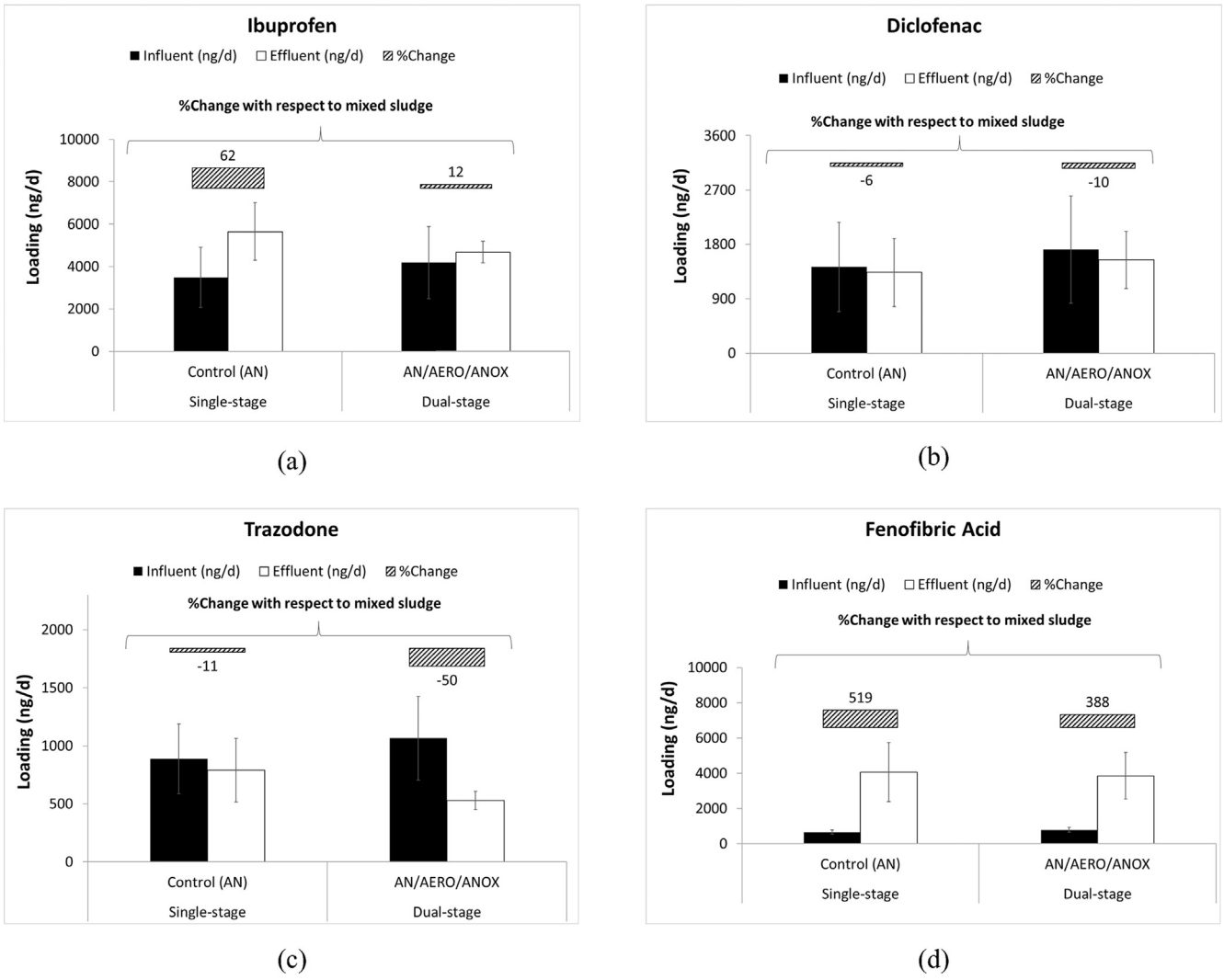


Fig. 5. Pharmaceuticals in single and dual-stage digesters**.** a. ibuprofen; b. diclofenac; c. trazodone; d. fenofibric acid.

erature reports high biodegradability for ibuprofen in general, the removal of ibuprofen in sequential sludge digestion is much lower than the removals reported for the main wastewater stream (Jelic et al., 2011; Langenhoff et al., 2013). This could be attributed to the absence of nitrifying conditions above 45°C in sludge digesters (Yuan et al., 2014). Alvarino et al. (2014) reported a linear relationship between ibuprofen degradation and nitrifying activity. In this study, since the 2nd stage AERO/ANOX sludge digester was operated at thermophilic conditions (55°C) with very limited to zero nitrifying activity, the removal efficiency of ibuprofen was much lower than the literature values.

Diclofenac removal increased from 6 to 10% in dual-stage digestion compared to single-stage conventional AN digestion, respectively (Fig. 5b). Diclofenac shows low and inconsistent removals (0–90%) in WWTPs (Jelic et al., 2012). The presence of chlorine atoms is believed to be linked with the low biodegradability of diclofenac (Kimura et al., 2005; Jelic et al., 2012). Literature also suggests the low efficiency (10–30%) of diclofenac in conventional WWTPs (Lindqvist et al. 2005; Joss et al. 2005). Physicochemical properties of diclofenac depicts its low removal efficiency. The affinity of a compound for a solid phase is represented by distribution constant (Kd). The distribution coefficient is the ratio between the concentration of a compound sorbed onto solids and concentration of the compound in liquid phase at equilibrium and it is determined experimentally. Studies reported that diclofenac showed a poor sorption onto sludge due to its low Kd of 16 L/kg suspended solids (SS), which is significantly lower than Kd of 500 L/kg SS required for significant sorption onto sludge (Ternes et al., 2004; Ternes, 2006). Likewise, another parameter used to determine the biodegradability potential of a compound is the kinetic constant (kbiol) expressed as L/g SS-d. Diclofenac has kbiol of 0.1 L/kg SS-d which is below the kbiol range of 0.1–10 L/g SS-d associated to partial removal of pharmaceuticals which makes it a very recalcitrant compound (Ternes, 2006).

Trazadone is used as an antidepressant. In single-stage AN digester trazodone showed only 11% removal while its removal was enhanced to 50% in sequential digestion as shown in Fig. 5c. The results are in accodance with the finding of Collado (2013). Collado collected samples from a municipal WWTP in Celra (Spain) receiving 80% of its influent flow from an industrial area with several pharmaceutical industries. The WWTP consisted of a conventional primary and secondary biological treatment followed by tertiary ultraviolet treatment (Collado, 2013). He found that trazodone showed medium removal efficiency (40–50%) between WWTP influent and secondary effluent samples and around 40–50% removal under ultraviolet radiation (Collado, 2013).

Fenofibric acid is used to lower low-density lipoprotien cholesterol also known as bad cholesterol. Fenofibric acid showed 519% accumulation in single-stage digester which was reduced to 388% accumulation in sequential AN/AERO/ANOX digestion (Fig. 5d). The results agree well with the findings of Rosal et al. (2010). He observed a very low removal (1.3%) of fenofibric acid in a treatment plant treating a mixture of municipal and industrial waste (Rosal et al., 2010).

5.2.2. Sterols

Sterols show a high removal efficiency (80–100%) in WWTPs (Furtula et al., 2011). Among the three sterols studied in this research, cholesterol was found to be most abundant while cholestanol was the least abundant in both influent and effluent samples of WWTPs (Furtula et al., 2011). Cholesterol showed 68 and 85% removal in single-stage and sequential digestion systems, respectively, as given in Fig. 6a.

While cholesterol was being removed, its degradation products, coprostanol and cholestanol were being formed. Both coprostanol and cholestanol showed 198 and 275% accumulation in the single-stage control digester as shown in Fig. 6b and c, respectively. The high accumulation of coprostanol and cholestanol in AN condition was because they can be produced naturally via AN diagenetic transformation of cholesterol (Chan et al., 1998). The accumulation of coprostanol and cholestanol was reduced to 39 and 76% respectively by the sequential AN/AERO/ANOX process.

5.2.3. PAHs

Biphenyl and C2-Phenanthrenes/Anthracenes were two PAHs studied in this research. Biphenyl is a compound that is used as a starting material in the production of polychlorinated biphenyls. Polychlorinated biphenyls are non-flammable and chemically stable and have high boiling points and excellent insulating properties (US EPA, n.d). PCBs are used in hundreds of industrial and commercial applications such as electrical equipment, heat transfer and hydraulic equipment (US EPA, n.d). In this study, biphenyl showed 15 and 59% removal in single-stage and sequential AN/AERO/ANOX digestion, respectively, as given in Fig. 7a.

Another PAH, C2-phenanthrenes/anthracenes, refers to a group of alkyl phenanthrene and anthracene compounds that have two carbon groups and contain three benzene rings (Mouwerik et al., 1997). In single-stage AN digestion, C2-phenanthrenes/anthracenes showed 12% accumulation while 24% removal was observed in sequential AN/AERO/ANOX digestion (Fig. 7b). The results are in accordance with the literature. Researchers conducted a sequential AN/AERO treatment study with upflow AN sludge blanket and AERO packed-bed biofilm reactors in-series to treat wastewater from petroleum refinery and achieved complete

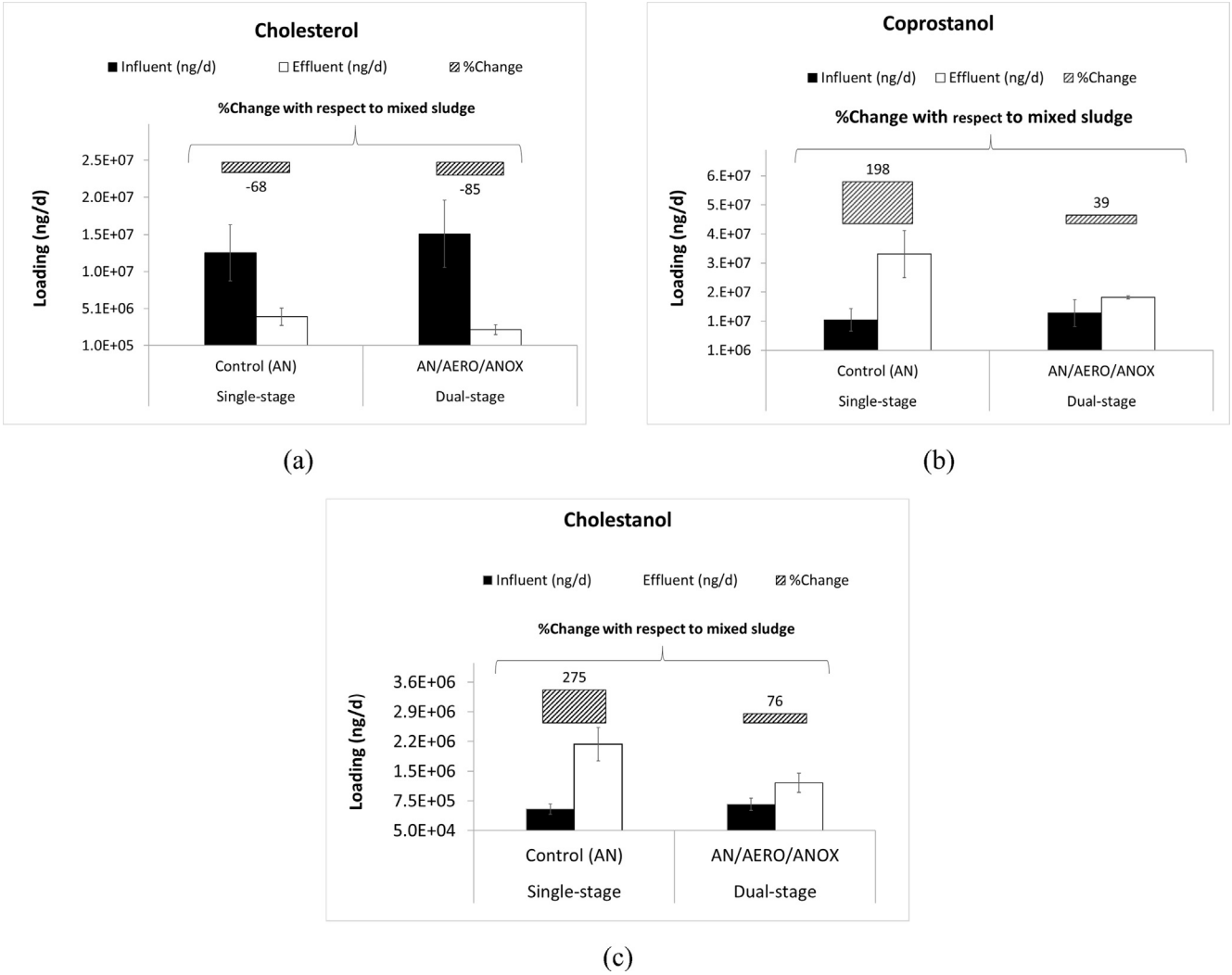
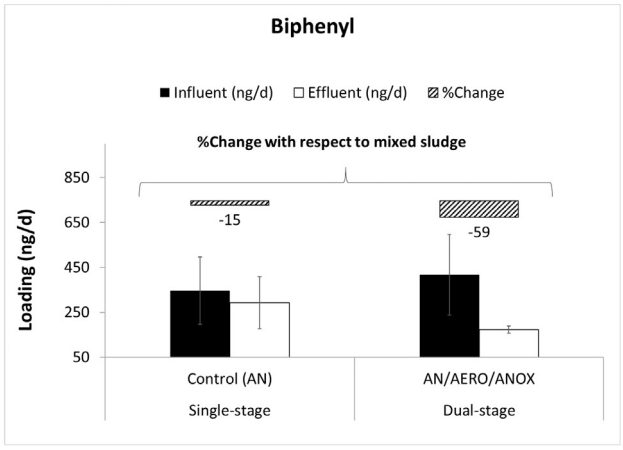
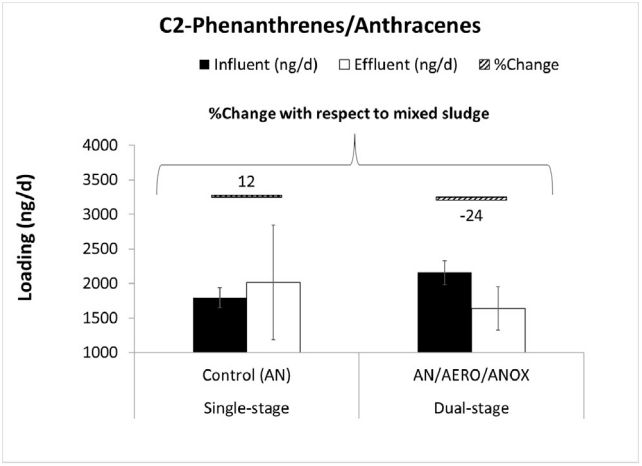


Fig. 6. Sterols in single and dual-stage digesters: a. cholesterol; b. coprostanol; c. cholestanol.



(a)



(b)

Fig. 7. Polycyclic aromatic hydrocarbons in single and dual-stage digesters: a. Biphenyl; b. C2-Phenanthrenes/Anthracenes.

removal of three PAHs (pyrene, phenanthrene and naphthalene) (Nasirpour et al., 2015)

6. Conclusions

The results of sequential AN/AERO/ANOX digestion demonstrated the advantage of using intermittent aeration in the 2nd stage digester treating AN effluent. Both single-stage (control) and sequential AN/AERO/ANOX digesters were operated under identical thermophilic conditions and overall SRT of 18 d. The following conclusions were drawn based on the experimental data and analysis;

- Overall, the 2nd stage AERO/ANOX digester not only reduced the effluent mass loading of all ECs by either reducing the accumulation or enhancing their removal but also provided enhancement in the removal of conventional pollutants. The removal of conventional pollutants (ammonia in particular) was higher in Scenario 1 with a 3 d-SRT compared to Scenario 2 with a 1.5 d-SRT, for the 2nd stage digester.
- Ibuprofen and fenofibric acid showed an accumulation tendency in both single-stage as well as sequential digestion systems. The rate of accumulation for ibuprofen was significantly higher (62%) in single-stage digestion compared to sequential digestion (12%). Similarly, sequential digestion decreased accumulation of fenofibric acid from 519% to 388%. Diclofenac and trazodone demonstrated 6 and 11% removal respectively in single stage digestion. These removals increased to 10 and 50% respectively in sequential AN/AERO/ANOX digestion.
- Sequential digestion also improved the removal of cholesterol from 68 to 85%. In contrast, degradation products of cholesterol (i.e., coprostanol and cholestenol) showed 198 and 275% accumulation in the single-stage digester, respectively. Both coprostanol and cholestanol demonstrated a decrease in accumulation to 39 and 76%, respectively, in sequential digestion.
- The removal of PAHs was also enhanced in sequential AN/AERO/ANOX digestion process. Biphenyl removal increased from 15 to 59% in sequential digestion compared to single-stage AN digestion. Likewise, 12% accumulation of C2-phenanthrene in single-stage digestion became 24% removal in the sequential digestion system.
- Overall, these results indicate that wastewater utilities using land application for biosolids management, in jurisdictions with restrictions on ECs along with pathogens and heavy metals, can utilize sequential AN/AERO/ANOX digestion to meet land application regulations and to reduce contamination of agricultural soil with ECs.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.wasman.2019.05.018.

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